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IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

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Foto: Dr. Matthias Otte

CONTENTS

MOTIVATION

NUCLEAR FUSION: ENERGY CHALLENGE

Lawson criterion **for net energy gain:**

$$n_\mathrm{e} T \tau_\mathrm{E} \geq \frac{12 f_\mathrm{tot}}{\langle \sigma_\mathrm{DT} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$
$$\gtrsim 3 \times 10^{21} \mathrm{keVs/m^3}$$

(requires temperatures $T \sim 10^1$-10^3 keV)

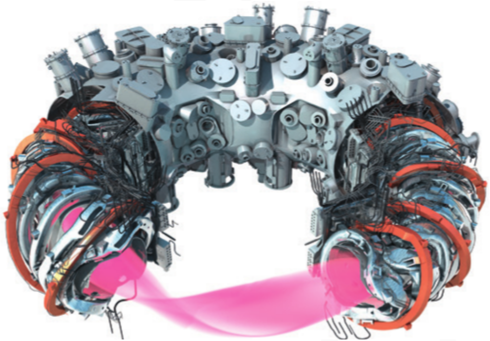
Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

WENDELSTEIN 7-X

Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma



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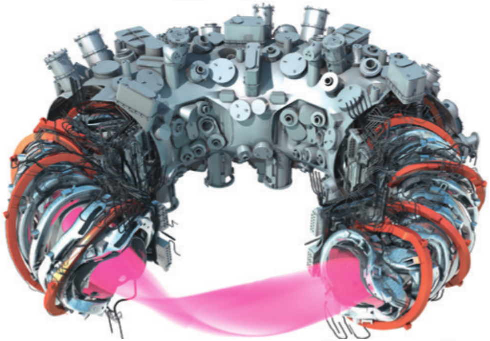
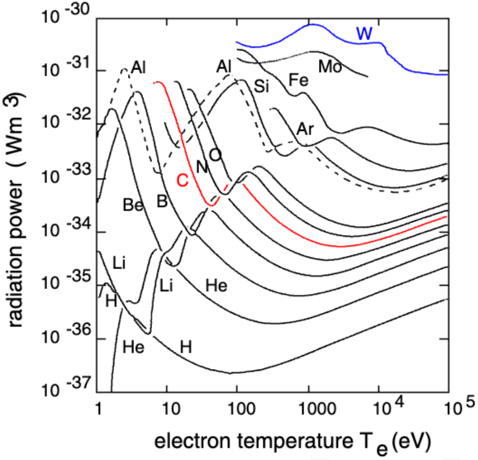


⇒ Dissipation of large fraction of heat load (energy) in scrape-off layer and particle flux to the target through *detachment*

PLASMA RADIATION

- adjust plasma profiles so radiation in SOL increases and transport across seperatrix reduced



EUROfusion IPP

PLASMA RADIATION

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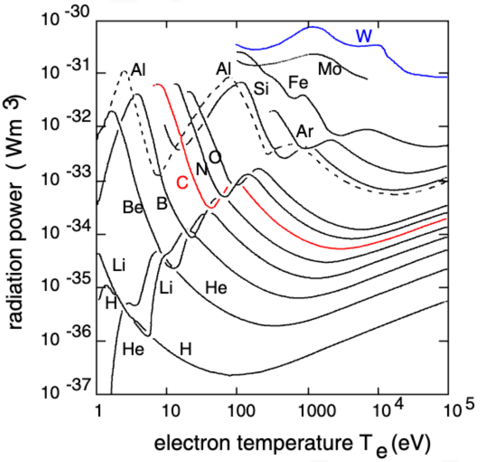
Impurity Seeding

Injection of low-Z (He, N_2) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling



PLASMA RADIATION

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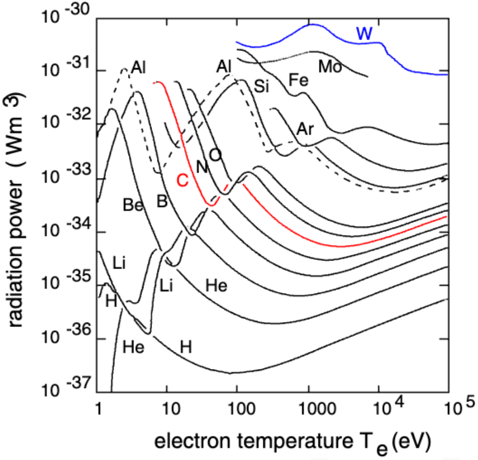
Impurity Seeding

Injection of low-Z (He, N_2) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling

- avoid critical impurity accumulation, use working gas

⇒ stable control w. radiation information?

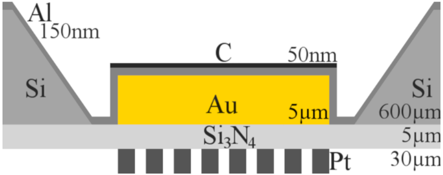


CORE BOLOMETER AT W7-X

BOLOMETER DIAGNOSTIC

Metal Resistor Bolometers:

- Thin metal (gold) film absorbers
- Wheatstone bridge circuit configuration
- Multicamera system: HBC & VBCl/r (32 + 2×24 channels)
- Spatial resolution: ~ 5 cm at magnetic axis



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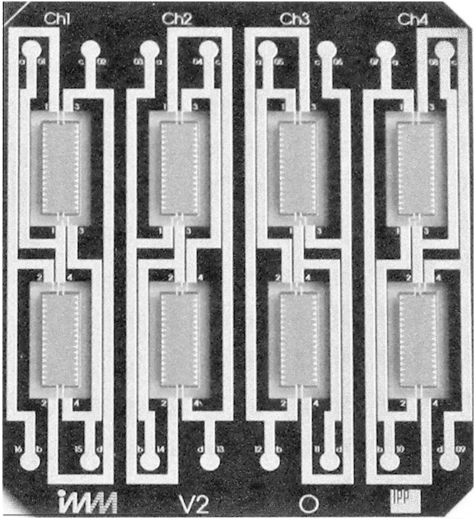

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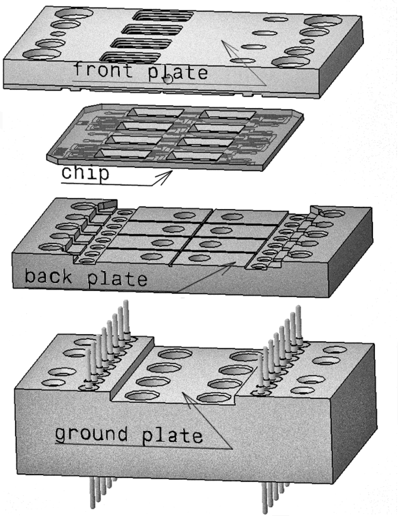

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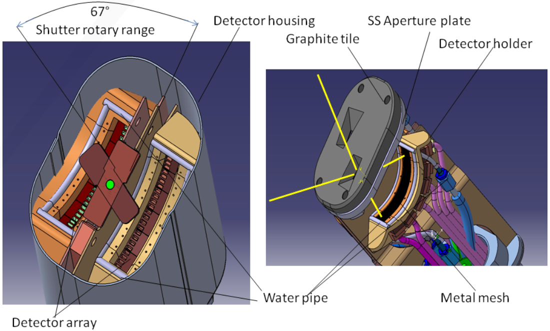
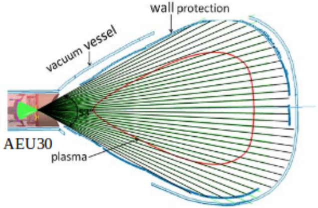
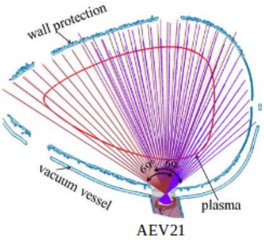




horizotal bolomter camera (HBC)

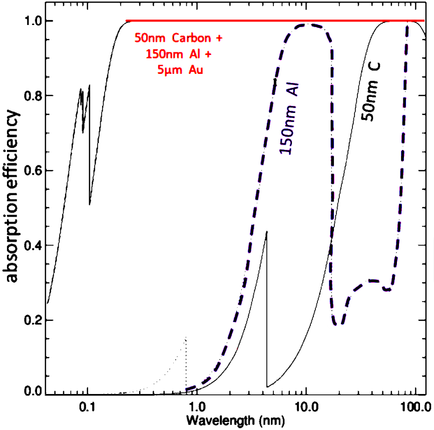


vertical bolometer camera (VBC)

PERFORMANCE

Performance Characteristics:

- Measures total radiation power (broadband)
- Near unity absorption efficiency $600\text{-}0.2\,$nm
- In-situ ohmic heating calibration with Wheatstone bridge $(F_{\mathrm{M}},\ f_{\mathrm{M}})$
- $\{0.4, 0.8, ...6.4\}$ ms sampling
- *Gaussian* uncertainty: $1.6\,\mu$W $\leftrightarrow$ SNR > 1000



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Power Per Detector:

$$P_M = F_M \cdot \left(\frac{d(\Delta \widetilde{U}_M)}{dt} + f_M \Delta \widetilde{U}_M \right)$$

Global Radiation Power:

$$P_\mathrm{rad} = \frac{V_\mathrm{P,tor}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

PERFORMANCE

Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ($P_{\mathrm{ch}} \sim P_{\mathrm{M}}/K_{\mathrm{M}}$)

Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

⇒ controlled cooling in the plasma edge

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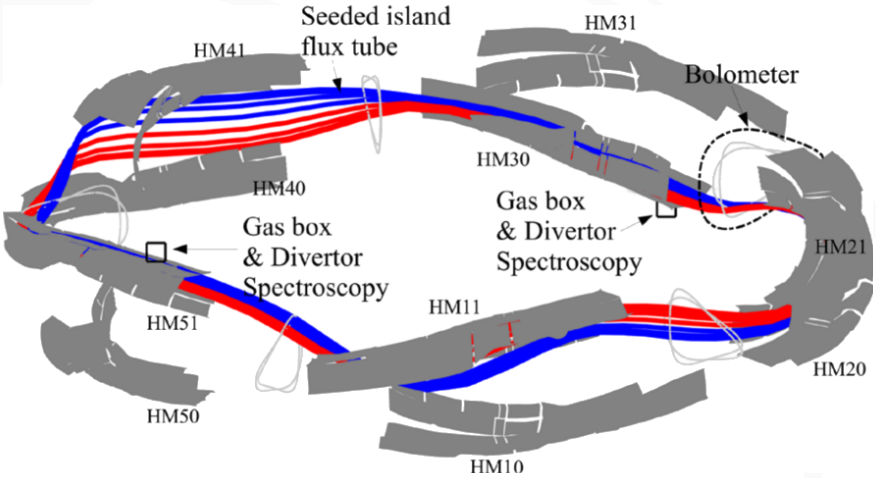
REAL-TIME RADIATION FEED-BACK CONTROL

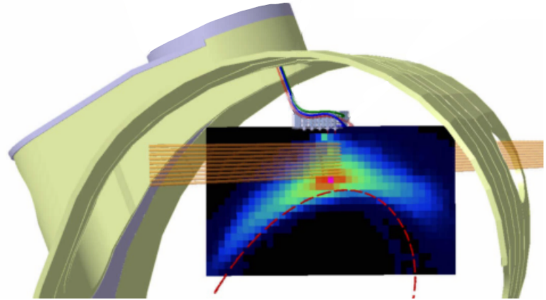
FEEDBACK SYSTEM



FEEDBACK SYSTEM

Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multi-diagnostic PID controller



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Radiation Prediction Proxies:

A: *Channel selection* S

$$P^{(1)}_{\text{pred}} = \frac{V_{\text{P,tor}}}{V_S} \sum_{\text{M}}^{S} \frac{P_M V_M}{K_M}$$

(e.g. $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$)

B: *Single channel*

$$P^{(2)}_{\text{pred}} \propto \Delta \widetilde{U_M}$$

(dimensionless, minimal overhead)

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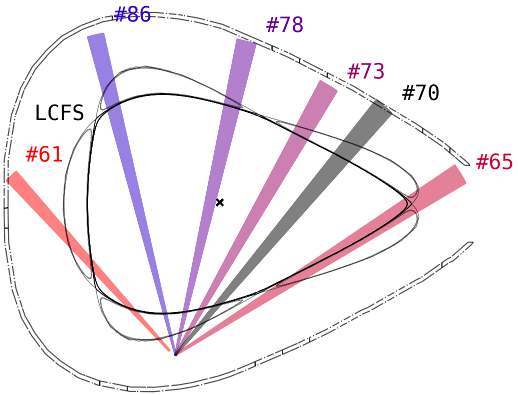
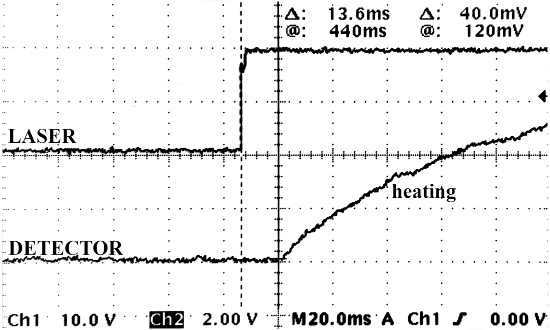


FEEDBACK SYSTEM

Performance:

- FIFO smoothing in $P^{(1)}_{\text{pred}}$: $\sim N\Delta t/2$
- Minimum acquisition latency $13.6\,\text{ms}$
- Total system latency: $150\text{-}400\,\text{ms}$ (dominated by plasma response)



Source: laboratory latency measurement with laser diode

EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Radiation fraction PID setpoint
  $f_{\mathrm{rad}} \sim 90\%$ off $P_{\mathrm{pred}}^{(1)}$
- 5-channel VBC subset
  $S = \{61, 65, 73, 78, 86\}$

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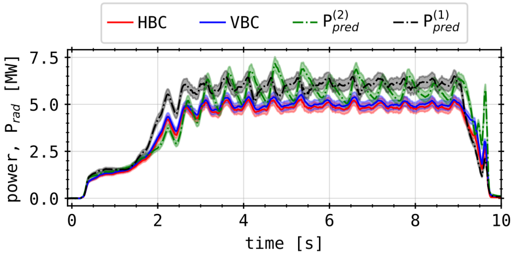

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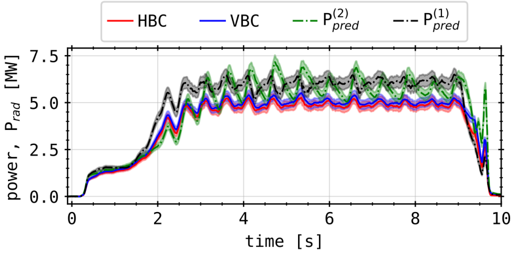

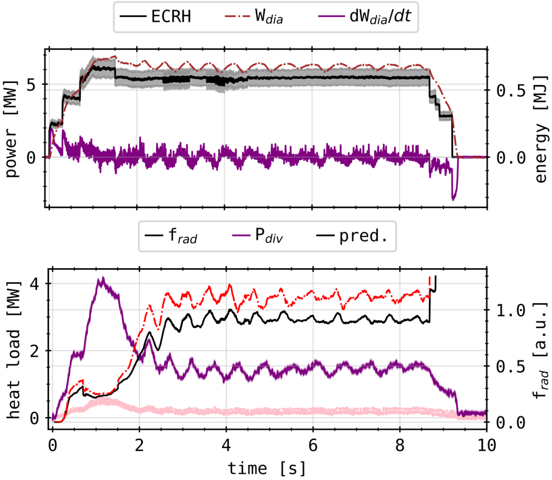

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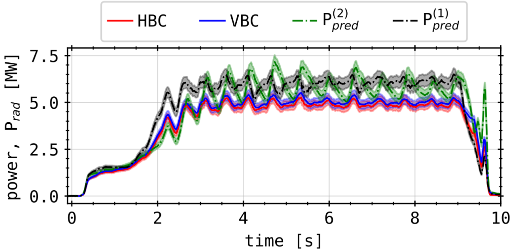
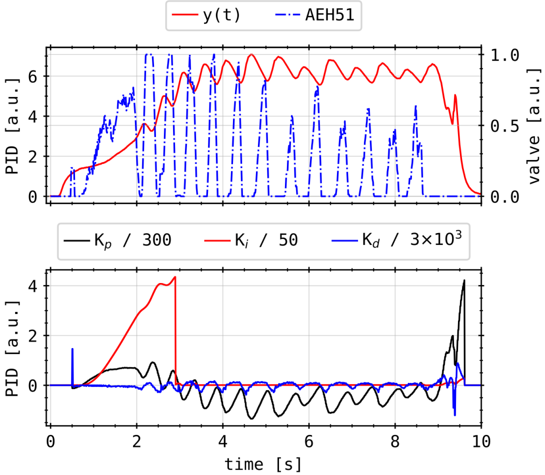
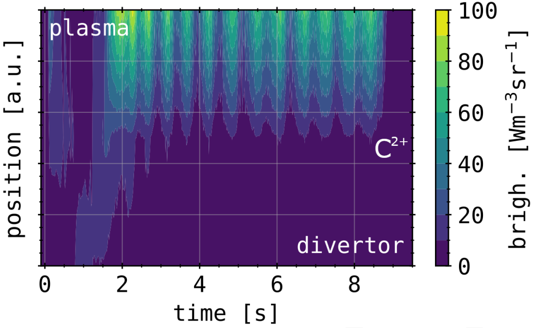




EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Peak $f_{\text{rad}} \gtrsim 100\%$, cycling with gas inlet
- Target heat load reduction factor ≥ 2
- C^{2+} detachment visible at $f_{\text{rad}} \sim 50\%$



Key Result

Demonstrated stable, real-time feedback-controlled radiative cooling with $f_{\text{rad}} \geq 85\%$ without terminal plasma disruption and significant target heat load reduction

LINE-OF-SIGHT SENSITIVITY

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- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
- Data set of all feedback discharges from OP1.2b
- Applied 5 different metric (linear & weighted deviation, FFT, ...)

Weighted deviation:

$$\varphi(t) = 1 - \frac{\left\| P_{\mathrm{pred}} - P_{\mathrm{rad}} \right\|(t)}{P_{\mathrm{rad}}(t)}$$

$$\vartheta = \frac{1}{T} \int_T \varphi(t)\,\mathrm{d}t$$

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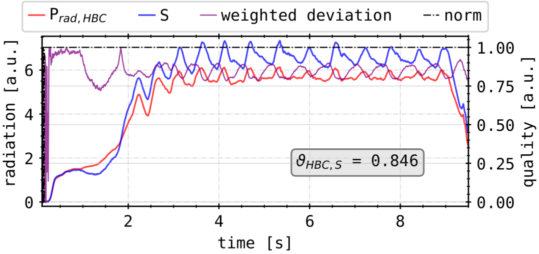
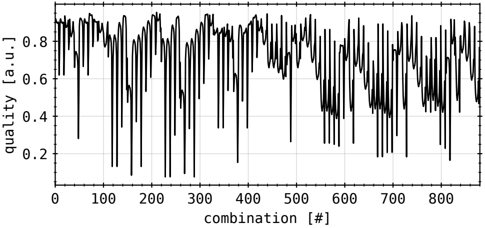
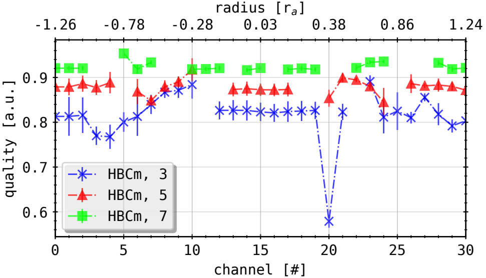


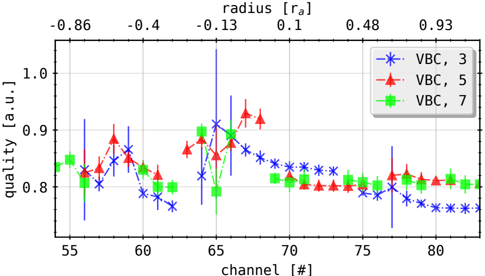


LINE-OF-SIGHT SENSITIVITY

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves ≥85% prediction accuracy
- VBC channels generally outperform HBC for edge sensitivity
- 3-5 channels sufficient for reliable P_{pred}





Takeaway

Detectors viewing close to separatrix and SOL most viable for feedback purposes

TOMOGRAPHY

MINIMUM FISHER REGULARIZATION

Measurement:

$$\mathbf{T}\,\vec{x} = \vec{b}$$

Bolometer tomography:

- 61 line-integrated measurements $\vec{b}$
- 2D emissivity $\vec{x}$: $\sim$4500 pixels
- *Geometry* $\mathbf{T}$ has high condition

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- 2D emissivity $\vec{x}$: $\sim$4500 pixels
- *Geometry* $\mathbf{T}$ has high condition
- ill-posed, requires regularization

Problem:

$$\min_{\eta} \left\| \left(\mathbf{T}^{\intercal}\mathbf{T} + \mu^{-1}\mathbf{H} \right)^{-1} \mathbf{T}^{\intercal}\vec{b} \right\| \rightarrow \vec{x}$$

Fisher Information:

$$I_{\mathrm{F}} = \int \frac{1}{x} \left(\frac{\partial x}{\partial \vec{r}} \right)^2 d\vec{r} \leftrightarrow \mathbf{H} \sim \frac{1}{x} \boldsymbol{\Delta}^{\intercal}\boldsymbol{\Delta}$$

limiting error on reconstruction:

$$\sigma_{\mathrm{x}} \leq \frac{1}{I_{\mathrm{F}}}$$

MINIMUM FISHER REGULARIZATION

Radially Dependent Anisotropy:

- Tailored weighting to poloidal gradient
- $k_{\mathrm{ani}} < 1$: favours anisotropy
- $k_{\mathrm{ani}} > 1$: favours smooth emissivity

$$\mathbf{H} \rightarrow \mathbf{H}_{\mathrm{ani}}$$

$$\boldsymbol{\Delta} = \boldsymbol{\nabla}_{\mathrm{r}}^{\mathsf{T}}(\cdot)\boldsymbol{\nabla}_{\mathrm{r}} + \boldsymbol{\nabla}_{\vartheta}^{\mathsf{T}}(\cdot)\boldsymbol{\nabla}_{\vartheta}$$

$$\boldsymbol{\nabla}_{\vartheta} \rightarrow \tilde{\boldsymbol{\nabla}}_{\vartheta} = \frac{1}{k_{\mathrm{ani}}\Delta r}\mathbf{D}_{\mathrm{dia}}^{(0)}\boldsymbol{\nabla}_{\vartheta}$$

MFR + RDA Advantages

- Robust to noisy data
- Incorporates a priori knowledge
- Stable convergence

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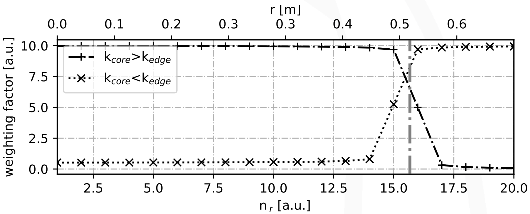

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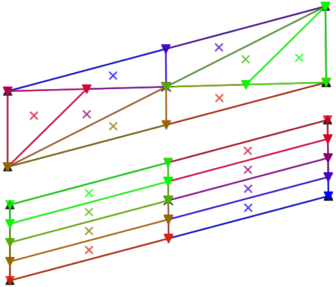
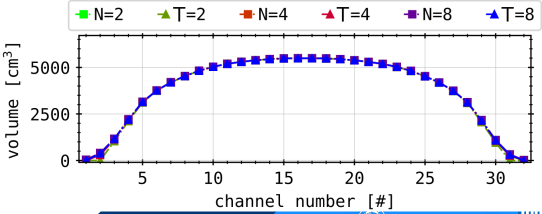
Geometry Perturbation Test:

- Previous and future calculations based on as-designed geometry





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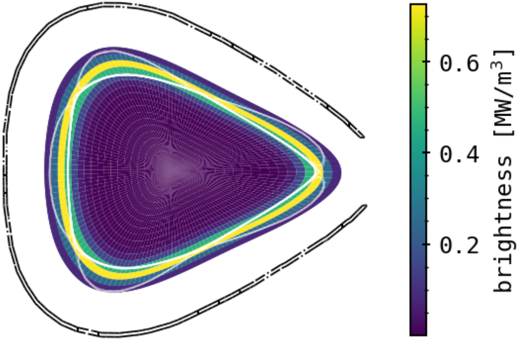
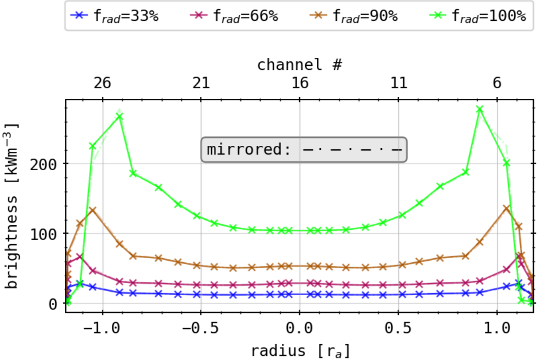


$(f_{\mathrm{rad}} \sim 100\%)$



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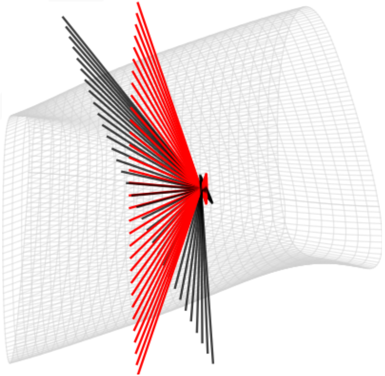

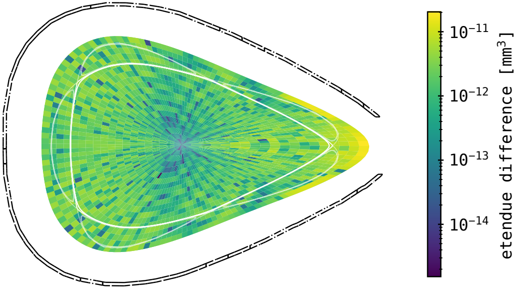

$$\|\mathbf{T} - \mathbf{T}_{\mathrm{sym}}\|$$

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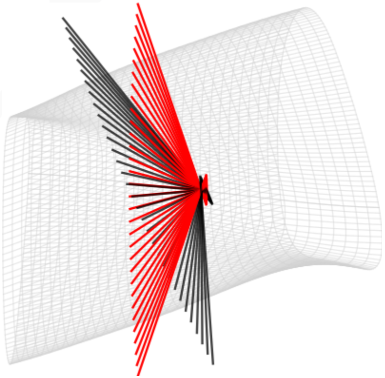

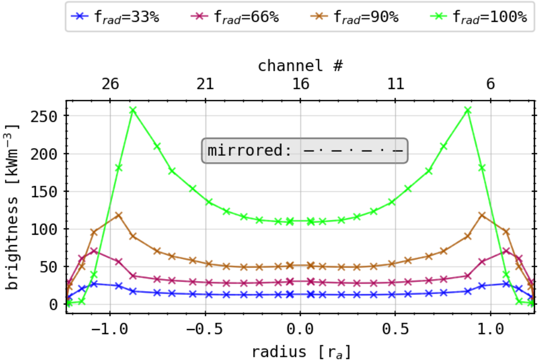

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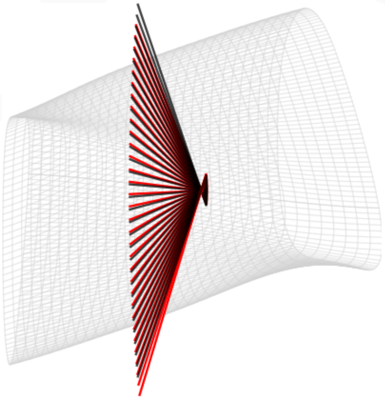

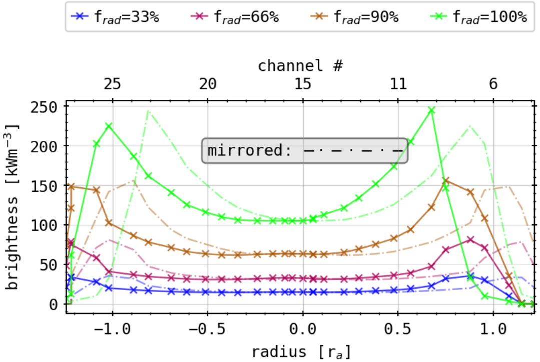

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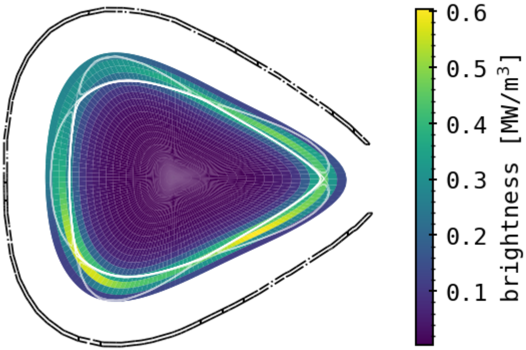
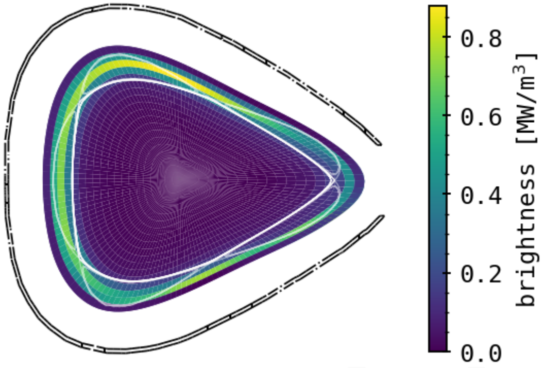


using $\mathbf{T}_{\mathrm{sym}} \cdot \vec{x} = \vec{b}$



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Results

- Discretization of system is computationally stable
- Intrinsic asymmetry of same order of magnitude as potential, unknown geometric pertubation - non-negligible
- Worst case machine warp scenario $\sim 0.2 r_\mathrm{a}$ shift

PHANTOM RECONSTRUCTIONS

Purpose

Regularized, parametric inversions need experienced tailoring for their respective application.

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Benchmarks:

- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
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Metrics:

$$\rho_{\mathrm{c}} = \frac{1}{\sigma_{pha}\sigma_{tom}} \mathrm{cov}\left(\vec{x}_{\mathrm{phan}}, \vec{x}_{\mathrm{tom}}\right)$$

$$M_{\mathrm{SD}} \sim \left\| \frac{\vec{x}_{\mathrm{tom}} - \vec{x}_{\mathrm{phan}}}{\left\| x_{\mathrm{phan}} \right\|} \cdot \vec{A}^{\mathsf{T}} \right\|$$

$$P^{\mathrm{2D}}_{\mathrm{rad,X}} = \vec{x}^{\mathsf{T}}_{\mathrm{X}} \cdot \vec{A}, \quad X = \{\mathrm{phan}, \mathrm{tom}\}$$

$$\chi^2 = \frac{1}{N_{\mathrm{ch}}} \left\| \left(\vec{b}_{\mathrm{phan}} - \vec{b}_{\mathrm{tom}}\right) \vec{\sigma}^{-1}_{\mathrm{bol}} \right\|$$

PHANTOM RECONSTRUCTIONS

Purpose

Regularized, parametric inversions need experienced tailoring for their respective application.

Benchmarks:

- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
- k_{ani} & phantom variations studies (radius, intensity etc.)
- Performed for only one magnetic grid configuration & LOS geometry

Metrics:

$$\rho_{\mathrm{c}} = \frac{1}{\sigma_{pha}\sigma_{tom}} \mathrm{cov}\left(\vec{x}_{\mathrm{phan}}, \vec{x}_{\mathrm{tom}}\right)$$

$$M_{\mathrm{SD}} \sim \left\| \frac{\vec{x}_{\mathrm{tom}} - \vec{x}_{\mathrm{phan}}}{\|x_{\mathrm{phan}}\|} \cdot \vec{A}^{\mathsf{T}} \right\|$$

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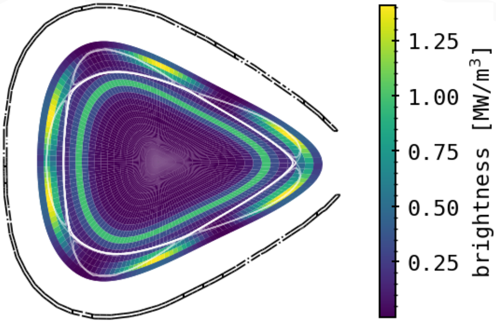

(Phantom: $\vec{b}_{\text{phan}}$)

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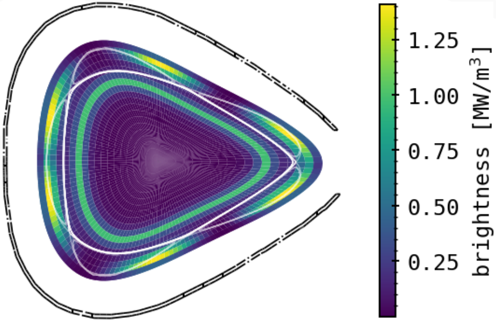

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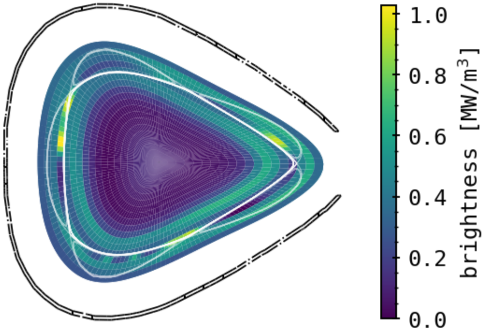

$k_{\mathrm{core}},\ k_{\mathrm{edge}} = \{2,\ 0.1\}$

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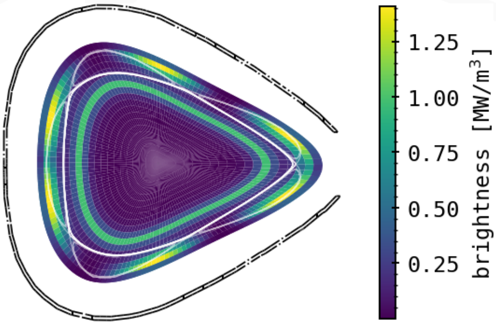

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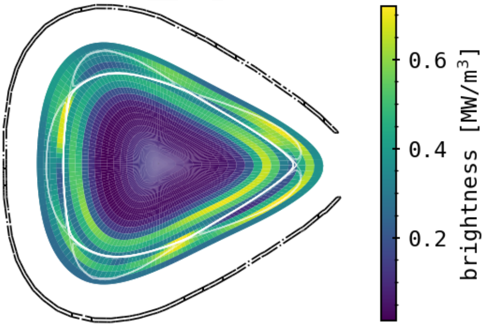
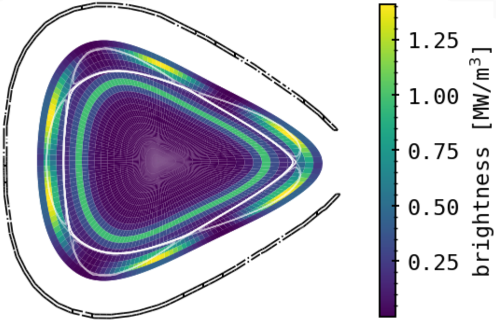


$k_{\text{core}},\ k_{\text{edge}} = \{2,\ 0.3\}$

PHANTOMS: EXAMPLE



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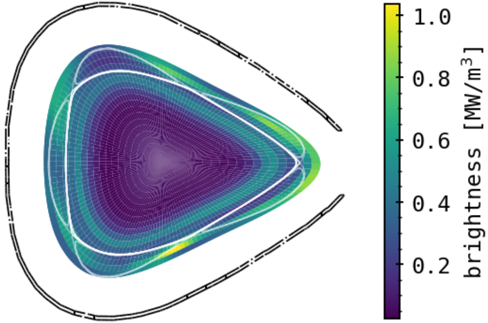

$k_{\mathrm{core}}, k_{\mathrm{edge}} = \{2,\, 0.6\}$

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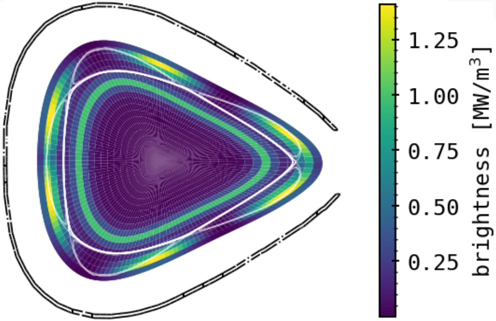

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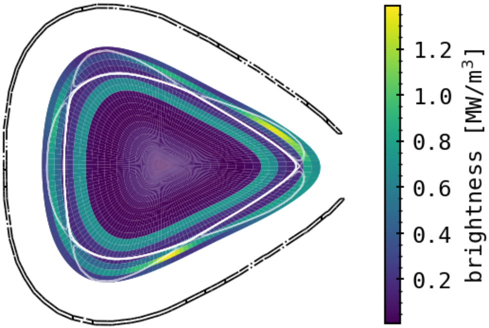

$k_{\text{core}},\ k_{\text{edge}} = \{20,\ 3.0\}$

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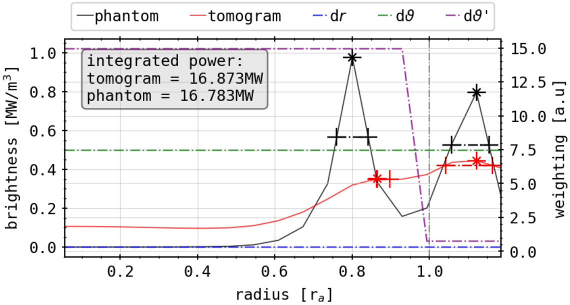

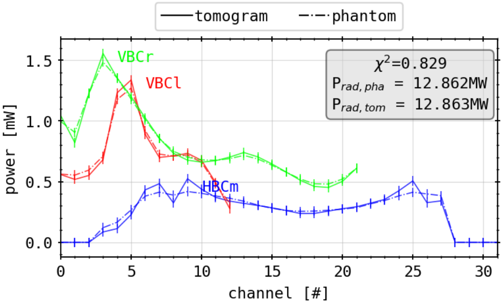

EUROfusion IPP

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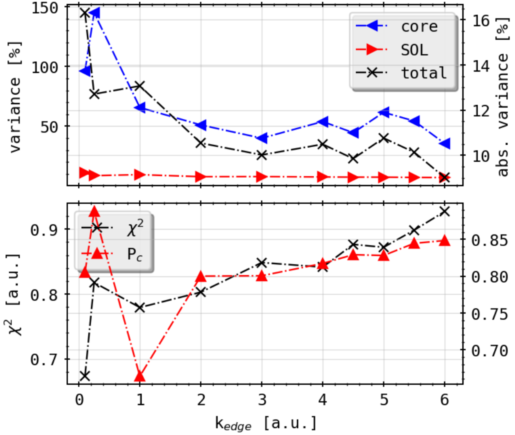
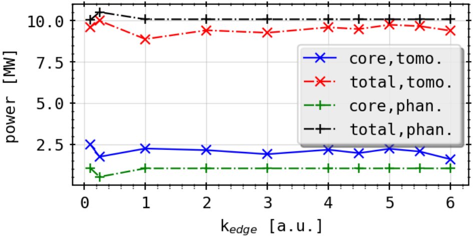
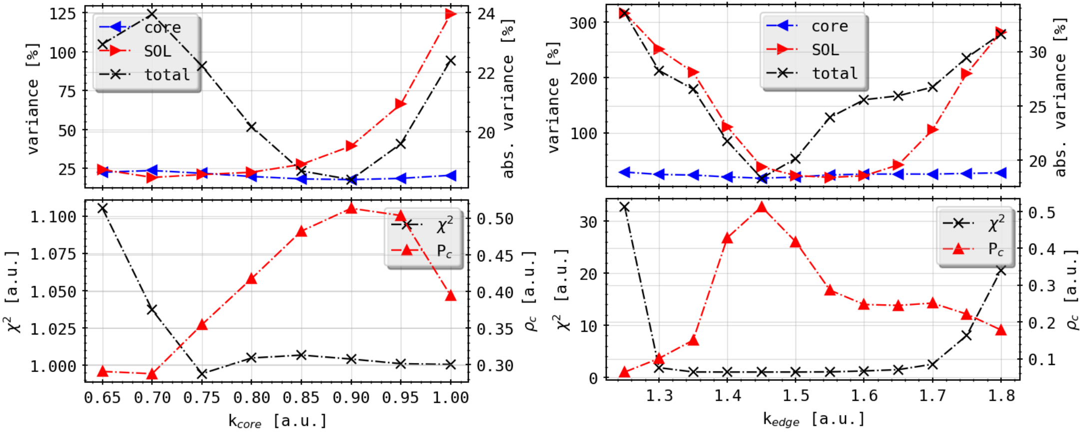




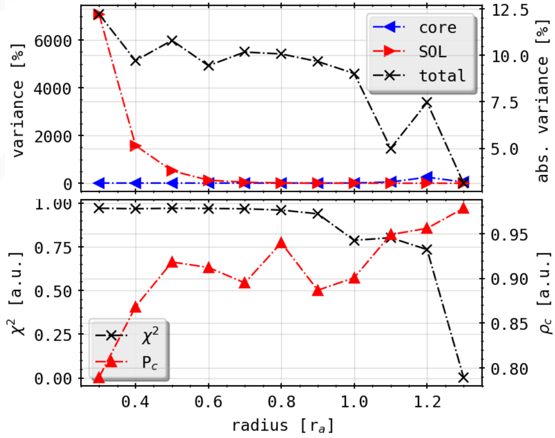
PHANTOMS: STATISTICS

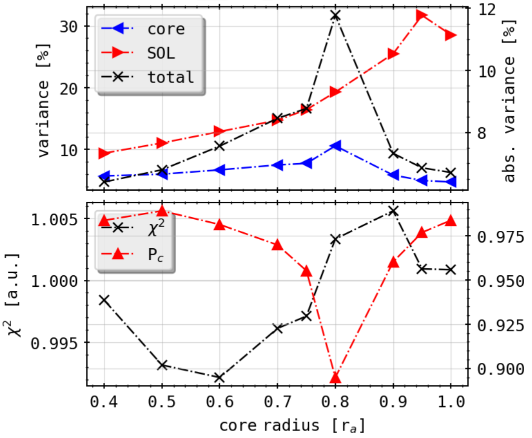
Isotropic Rings: k_{ani} scan



PHANTOMS: STATISTICS

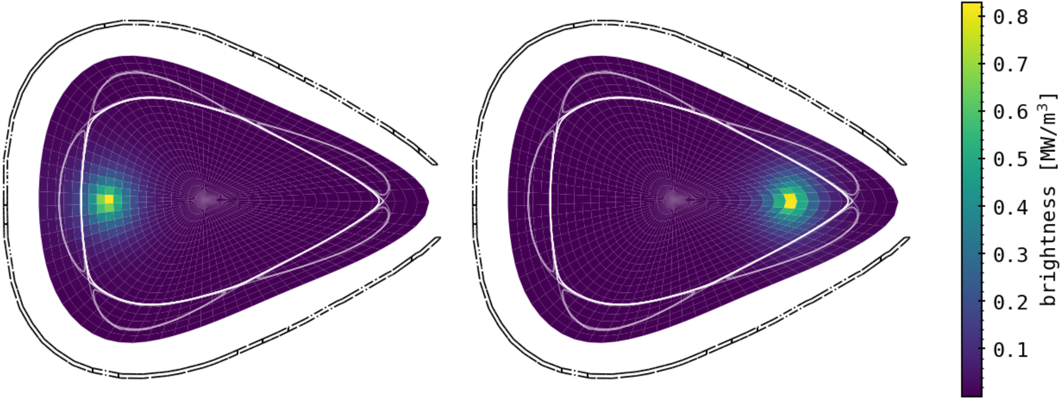
Isotropic Rings: Radial Scan





PHANTOMS: STATISTICS

Composite Distributions



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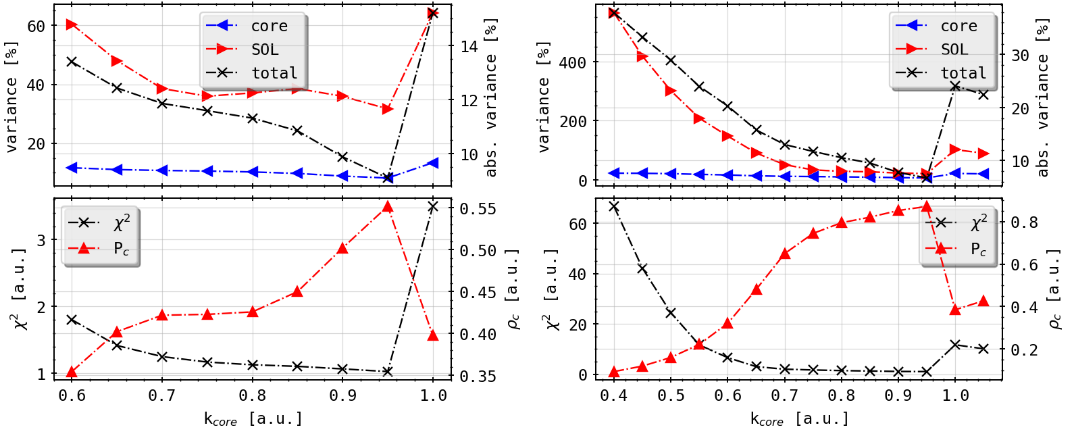
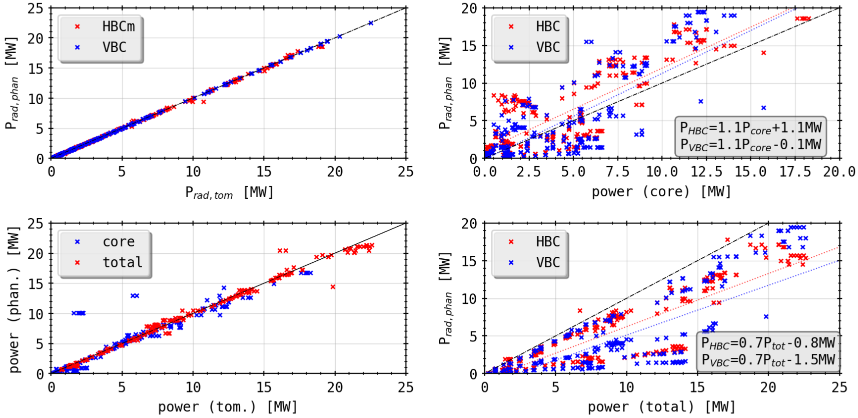


PHANTOMS: STATISTICS

Full Set



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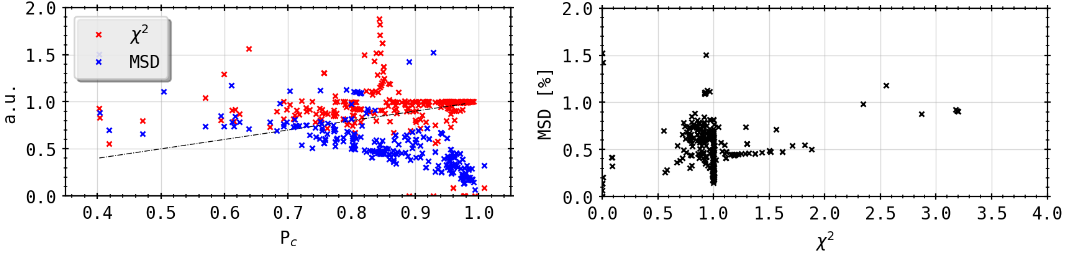

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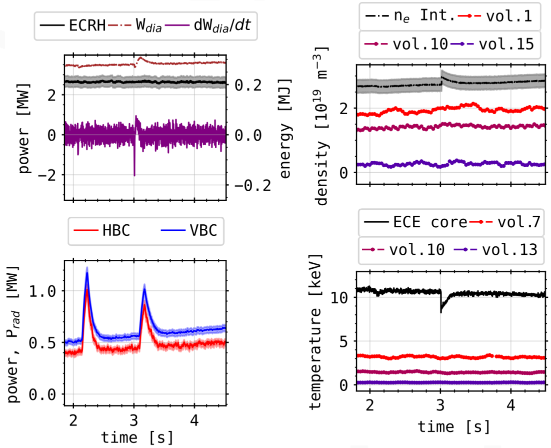
Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for physics analysis and power balance calculations, though computational limitations prevent continuous real-time operation

EXPERIMENTAL RECONSTRUCTIONS

Example: XP20180809.13



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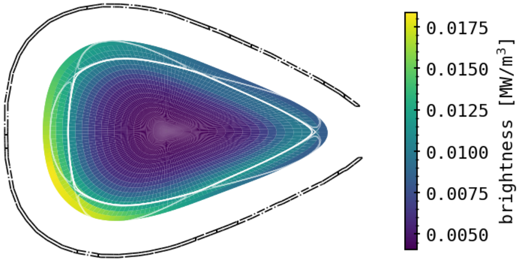

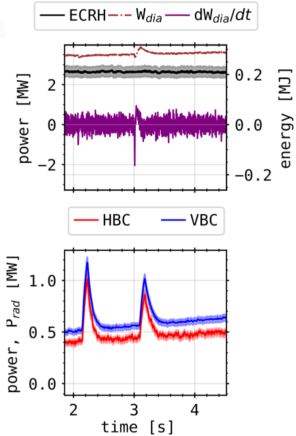

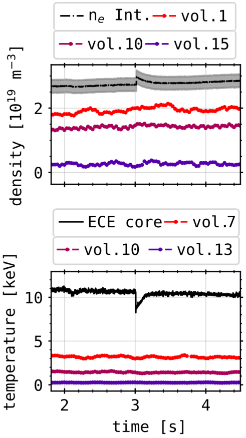

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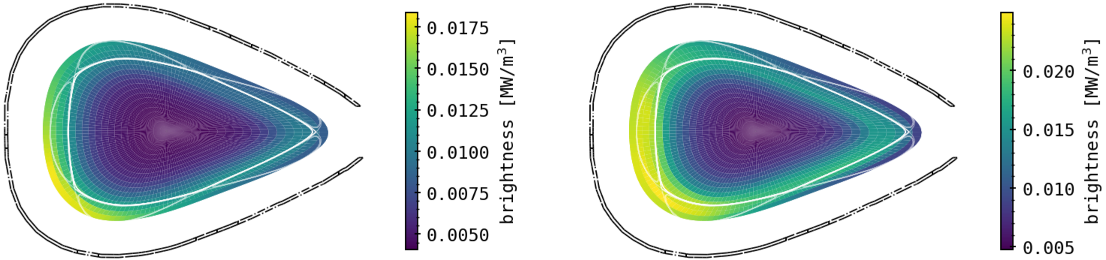

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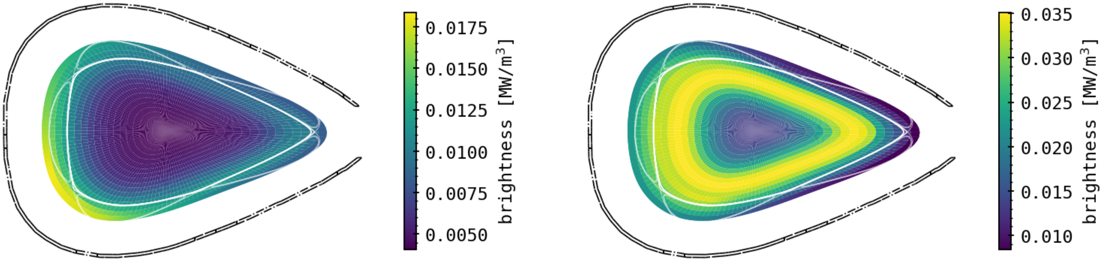

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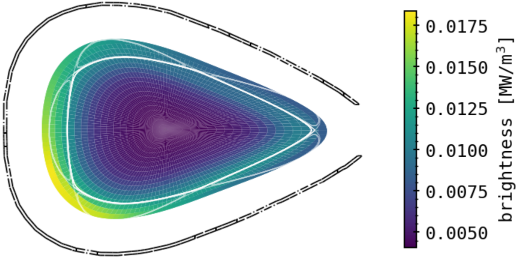
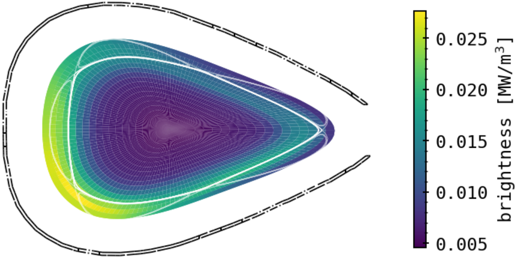
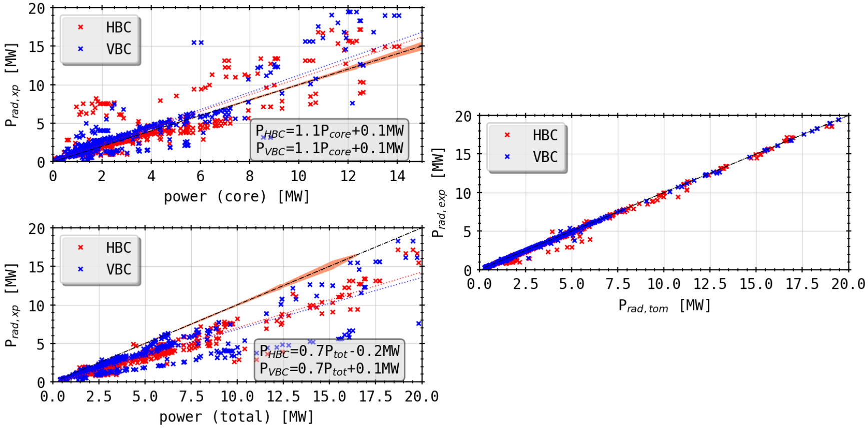




EUROfusion IPP

EXPERIMENTAL RECONSTRUCTIONS

Statistics: Full Set



GLOBAL POWER BALANCE

Simplified 0D Ansatz:

$$
\begin{aligned}
0 &\stackrel{!}{=} S + \frac{1}{V_\mathrm{P}} \int \frac{3}{2} \left(\frac{\partial p}{\partial t} \right) + p \nabla \vec{v} + \nabla \vec{q} \, \mathrm{d}\vec{r} \\
&= - P_\mathrm{H} + P_\mathrm{rad} + P_\mathrm{div} + \frac{\mathrm{d}W}{\mathrm{d}t} \\
&= P_\mathrm{bal}
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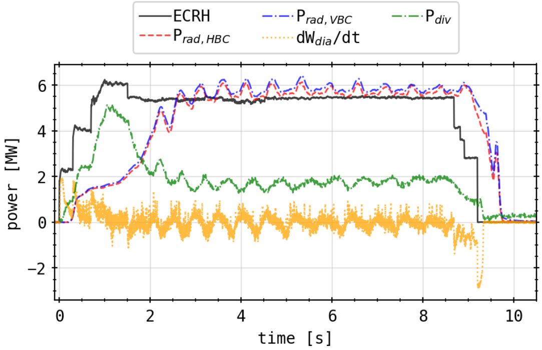

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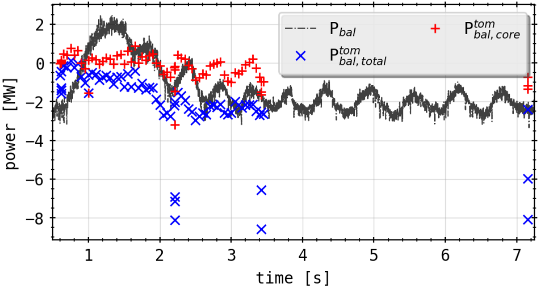

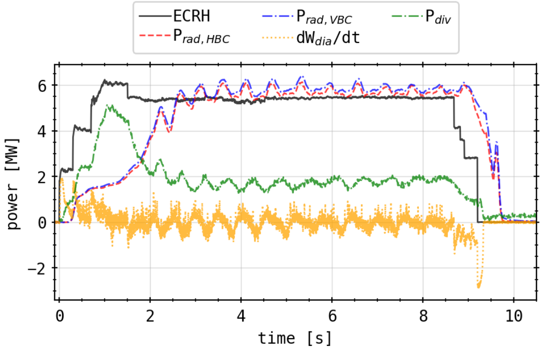

EUROfusion IPP

Statistical Analysis:

- Forward vs backward P_{rad} correlation
- Tomographic integral: $P_{\text{rad,tomo}} = \iiint \epsilon(\mathbf{r})\, dV$
- Agreement with bolometer sum within 10-20%
- Core radiation slightly overestimated
- Reveals SOL radiation underestimation in some cases
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Power Balance Application:

- P_{2D}^{core} improves balance accuracy
- More stable than P_{rad} alone
- Alternative for quasi-steady-state analysis

CONCLUSIONS AND OUTLOOK

KEY ACHIEVEMENTS

1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
2. **Line of Sight Optimization:** Systematic sensitivity analysis identified optimal detector configurations
3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting

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1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
   - Achieved stable detachment with $f_{\mathrm{rad}} \geq 85\%$, peaks up to 100%
   - Reduced target heat loads by factor >2
   - Minimum system latency: $13.6\,\mathrm{ms} + 200 - 400\,\mathrm{ms}$ plasma response
   - Validated 3-5 channel LOS subsets for fast prediction ($\geq 85\%$ accuracy)

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3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting
   - Robust to geometry perturbations ($\pm 1^\circ$) and noise (10%)
   - Successfully applied to experimental data (OP1.2b)
   - Reconstruction quality $>$90% for well-constrained phantoms
   - Provides alternative for power balance calculations (10-20% agreement)

SCIENTIFIC INSIGHTS

Detachment Physics:

- C^{3+} detachment visible at $f_{\mathrm{rad}} \sim 50\%$
- Stable detachment achievable at $f_{\mathrm{rad}} > 90\%$
- Radiation shifts inward as $f_{\mathrm{rad}} \Rightarrow 1$
- X-point and island radiation concentration

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Diagnostic Performance:

- Multicamera system provides comprehensive coverage
- Geometry uncertainties affect results
- Tomography reveals SOL radiation underestimation

FUTURE OUTLOOK AND RECOMMENDATIONS

System Upgrades:

- Reduce feedback latency: target <10 ms
- Implement more complex P_{pred} models (machine learning)
- Per-experiment configurability and auto-tuning
- Develop predictive scaling laws for gas injection
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- Real-time capable algorithms (GPU acceleration)
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- Automated parameter optimization (grid search, ML)
- Integration with synthetic diagnostics

Reactor Relevance:

- Essential for DEMO power plant operation (P_{fus} $\sim$500 MW)

BROADER IMPACT

Contribution to Fusion Energy Research

This thesis advances the state-of-the-art in plasma diagnostics and control for stellarator fusion devices, providing essential tools and knowledge for achieving reactor-relevant operating scenarios with controlled radiative cooling and stable detachment.

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- Validated diagnostic optimization methods
- Benchmarked tomography algorithm
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Scientific Contributions:

- Detachment physics understanding
- Impurity transport insights
- Radiation distribution characterization
- Power balance methodology

SUMMARY

- Developed and operated first real-time bolometer radiation feedback system at W7-X stellarator
- Achieved stable, controlled detachment with radiation fractions $f_{\mathrm{rad}} \geq 85\%$ and peaks up to 100%
- Reduced divertor target heat loads by factor of 2 through controlled impurity seeding
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- Validated 3-channel bolometer subset for fast radiation power prediction with $\geq$85% accuracy
- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback
- STRAHL modeling revealed carbon dominance in SOL radiation and inward shift at high f_{rad}
- Developed and benchmarked Minimum Fisher Regularization tomography with radially dependent anisotropy
- Demonstrated robust 2D radiation reconstruction from experimental data
- Provided alternative power balance methodology using tomographic integrals
- Established essential approach for future fusion reactor heat load management

COLLABORATIONS

Acknowledgement

This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Collaborators:

Wendelstein 7-X

IPP

HELMHOLTZ
RESEARCH FOR GRAND CHALLENGES

EUROfusion

Thank You

REFERENCES I

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STRAHL Code:

- 1D impurity transport and radiation simulation
- Solves radial continuity for each ionisation stage
- Experimental input data: pertubation study in transport coefficients, profile smoothing, source model etc.
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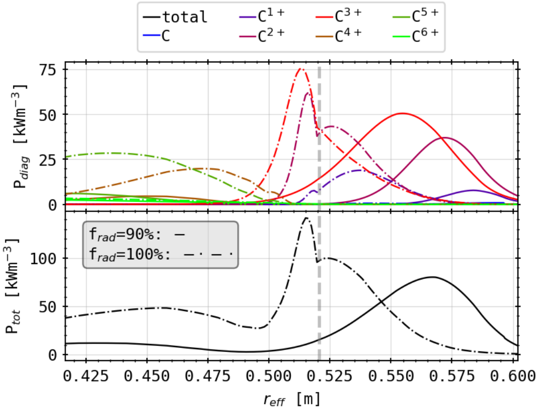

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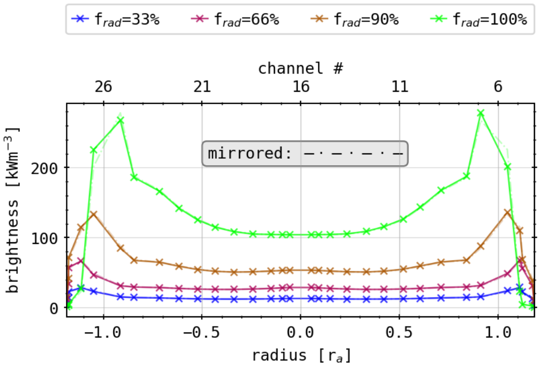
Key Results:

- Carbon dominates emissivity ($C^{2/3+}$, x10^2+ of Oxygen levels)
- Inward shift of radiation for $f_{\mathrm{rad}} \to 1$ toward $r \sim 0.95 r_{\mathrm{a}}$
- Radiation moves from outside to inside separatrix

STRAHL IMPURITY TRANSPORT MODELING



Forward Modeling:

- Chord brightness from STRAHL results
- Compare with experimental measurements
- Validate assumptions about impurity radiation and transport

Discrepancy

Forward calculations show asymmetry - less than experiments - in confinement, suggesting intrinsic geometry effects

STRAHL IMPURITY TRANSPORT MODELING